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## ALIGNING VERTEBRAL BODIES


### Abstract

Misaligned bones on opposite sides of a joint are aligned using a first rigid extension securable to one of the misaligned bones using a particular surgical approach, and a second rigid extension having a contacting surface positionable in contact with the other the two misaligned bones from the same surgical approach. The first and second rigid extensions are moved with respect to each other using a lever, whereby a pulling force is exerted on one of the bones, and a pushing force on the other, thereby aligning the first and second misaligned bones.



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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. patent application Ser. No. 18/321,966, filed on May 23, 2023 and published as US 2024-0325064, which is a continuation of U.S. patent application Ser. No. 16/739,424, filed on Jan. 10, 2020 (published as U.S. 2020-0214753), which is a continuation of U.S. patent application Ser. No. 13/483,558, filed May 30, 2012 (abandoned), all of which are incorporated by reference herein in their entireties for all purposes.

FIELD
[0002] The invention relates to aligning bones of the body, and more particularly to reducing adjacent vertebrae and realigning the vertebral column, for example to correct spondylolisthesis.
BACKGROUND
[0003] Bones, bony structures and tissue are susceptible to a variety of weaknesses that can affect their ability to provide support and structure. Weaknesses in bony structures may have many causes, including degenerative diseases, tumors, fractures, and dislocations. Advances in medicine and engineering have provided doctors with a plurality of devices and techniques for alleviating or curing these weaknesses.
[0004] In some cases, an anterior or posterior displacement of a vertebra with respect to an adjacent vertebra takes place, giving rise to a disorder termed spondylolisthesis. Resultant pressure on nerves can produce pain, stiffening of the back, changes in gait, and muscular atrophy, for example.
[0005] Conservative approaches may include physical therapy and treatment with NSAIDs such as acetaminophen. Spinal decompression through, for example, laminectomy or non-surgical methods may be employed. Additionally, a fusion may be carried out, for example a posterolateral fusion of adjacent vertebrae.
SUMMARY
[0006] In accordance with the disclosure, a device for therapeutically aligning two misaligned vertebral bodies on opposite sides of a joint, comprises a first rigid extension securable to a first of the misaligned bones using a single approach selected from the group consisting of left lateral

approach, right lateral approach, anterior approach, and posterior approach; a second rigid extension having a contacting surface positionable in contact with the second of the two misaligned bones from the approach selected for securing the first rigid extension, the second rigid extension moveable with respect to the first rigid extension when the first rigid extension is secured to the first of the misaligned bones and the second rigid extension is positioned in contact with the second of the two misaligned bones; and a lever configured to lever the first rigid extension with respect to the second rigid extension to move the first rigid extension with respect to the second rigid extension, to thereby exert a pulling force on the first of the two misaligned bones, and a pushing force on the second of the two misaligned bones, aligning the first and second misaligned bones.
[0007] In embodiments thereof, the first and second misaligned bones are vertebrae; the device further includes an intervertebral spacer, the first rigid extension securable to the intervertebral spacer, the intervertebral spacer connected to the first of the misaligned bones; and the intervertebral spacer is formed in two segments, each segment attachable to one of the first or second misaligned bones, the segments moveable with respect to each other when attached to a respective misaligned bone.
[0008] In other embodiments thereof, the segments moveable in ratcheting engagement with respect to each other; the lever applies leverage using a rotatable thread; the first rigid extension is securable to the misaligned bone using a threaded fastener; the first rigid extension is a plate; the plate forms a levered connection between the first and second misaligned bones; the first rigid extension extends around the first misaligned bone to contact a portion of the first misaligned bone which faces an approach different from the approach selected; and one of the first or second misaligned bones is the sacrum.
[0009] In yet further embodiments thereof, the lever is a screw having threads configured to engage bone on a first end, and threads configured to engage a nut on a second, opposite end, the nut rotated about the second end moves the second rigid extension and to thereby lever the first and second misaligned bones into alignment; the device further includes an intervertebral spacer; and the intervertebral spacer is expandable to increase a distance between the first and second misaligned bones.
[0010] In another embodiment of the disclosure, a device for therapeutically aligning two vertebrae, comprises a first rigid extension securable to a first of the vertebrae using a single approach selected from the group consisting of left lateral approach, right lateral approach, anterior approach, and posterior approach; a second rigid extension having a contacting surface positionable in contact with the second of the two vertebrae from the approach selected for securing the first rigid extension, the second rigid extension moveable with respect to the first rigid extension when the first rigid extension is secured to the first of the vertebrae and the second rigid extension is positioned in contact with the second of the two vertebrae; a lever configured to lever the first rigid extension with respect to the second rigid extension to move the first rigid extension with respect to the second rigid extension, to thereby exert a pulling force on the first of the two vertebrae, and a pushing force on the second of the two vertebrae, aligning the first and second vertebrae; and a fastener configured to maintain the first and second vertebrae in relative alignment.
[0011] In embodiments thereof, the device further includes an intervertebral spacer connectable to at least one of the first and second rigid extensions during alignment of the vertebrae, and connectable to both vertebrae to maintain the vertebrae in relative alignment; and the first rigid extension forming a first portion of an intervertebral spacer, the second rigid extension forming a second portion of an intervertebral spacer, the first and second portions mateably engageable and maintainable in relative mutual engagement using a ratchet.
[0012] In a further embodiment of the disclosure, a method for therapeutically aligning two misaligned bones on opposite sides of a joint, comprises securing a first rigid extension to a first of the misaligned bones using a single approach selected from the group consisting of left lateral approach, right lateral approach, anterior approach, and posterior approach; contacting the second

of the misaligned bones with a second rigid extension, the second rigid extension having a contacting surface positionable in contact with the second of the two misaligned bones from the approach selected for securing the first rigid extension, the second rigid extension moveable with respect to the first rigid extension when the first rigid extension is secured to the first of the misaligned bones and the second rigid extension is positioned in contact with the second of the two misaligned bones; and levering the first rigid extension with respect to the second rigid extension using a lever configured to move the first rigid extension with respect to the second rigid extension, to thereby exert a pulling force on the first of the two misaligned bones, and a pushing force on the second of the two misaligned bones, aligning the first and second misaligned bones.
[0013] In embodiments thereof, the method further includes inserting an intervertebral spacer between the two misaligned bones; and further includes connecting the intervertebral spacer to the first of the misaligned bones; and applying the lever between the intervertebral spacer and the second of the misaligned bones.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:
[0015] FIG. **1** is a diagram of a joint of a body;
[0016] FIG. **2** is a diagram of the joint of FIG. **1**, misaligned;
[0017] FIG. **3** is a diagram of the joint of FIG. **1**, and a cross-section of a connected plate and spacer of the disclosure;
[0018] FIG. **4** depicts the joint of FIG. **1**, aligned using the plate and spacer of FIG. **3**;
[0019] FIG. **5**A depicts the joint of FIG. **1**, and a cross section of a spacer, plate, and tool of the disclosure
[0020] FIG. **5**B is a diagram of the joint of FIG. **1**, the joint having been aligned using the spacer, plate, and tool of FIG. **5**A;
[0021] FIG. **6** is a diagram of the joint of FIG. **1**, and the spacer and plate of FIG. **5**B secured to the joint using fasteners, from a cross-section;
[0022] FIG. **7** is a diagram of the joint of FIG. **1**, and a top view of a reaching tool and plate of the disclosure;
[0023] FIG. **8** depicts the joint and device of FIG. **7**, in a cross-section;
[0024] FIG. **9** depicts the joint and device of FIG. **8**, the joint aligned using the device of FIG. **8**;
[0025] FIG. **10** is a diagram of the joint of FIG. **1**, and a cross-section of spacer of the disclosure having two ratcheting segments;
[0026] FIG. **11** depicts the joint and device of FIG. **10**, the joint aligned by the device of FIG. **10**;
[0027] FIG. **12** is a cross-section of a segmented spacer of the disclosure;
[0028] FIG. **13** depicts the spacer of FIG. **12**, the segments mutually secured by a fastener;
[0029] FIG. **14** is a diagram of the joint of FIG. **1**, a cross-section of a ratcheting spacer of the disclosure, and an indication of an application of a force to align the joint;
[0030] FIG. **15** is a diagram of the joint of FIG. **1**, and a spacer of the disclosure, and an indication of an application of a force to align the joint;
[0031] FIG. **16** is a diagram of the joint of FIG. **1**, and a cross-section of a spacer and tool in accordance with the disclosure;
[0032] FIG. **17** depicts the joint and device of FIG. **16**, the device having aligned the joint;
[0033] FIG. **18** depicts the joint and spacer of FIG. **17**, the spacer secured to the aligned joint;
[0034] FIG. **19** is a diagram of the joint of FIG. **1**, and a cross-section of a spacer and tool of the

disclosure;
[0035] FIG. **20** is a diagram of the joint of FIG. **1**, and a reaching plate of the disclosure;
[0036] FIG. **21** is a diagram of the joint of FIG. **1**, and a hinged plate and lever of the disclosure;
[0037] FIG. **22** is a diagram of the joint of FIG. **1**, and a segmented spacer and lever of the disclosure;
[0038] FIG. **23** is a diagram of the joint and device of FIG. **22**, the joint having been aligned by applying the lever to the device;
[0039] FIG. **24** depicts the joint and device of FIG. **23**, the joint having been aligned by applying the lever to a segment and to the joint;
[0040] FIG. **25** depicts the joint and device of FIG. **24**, the segments joined by a fastener;
[0041] FIG. **26** is a diagram of the joint of FIG. **1**, and a cross-section of a spacer and plate of the disclosure;
[0042] FIG. **27** depicts the joint and spacer of FIG. **26**, the joint having been aligned by the plate, and the spacer secured to both sides of the joint using fasteners;
[0043] FIG. **28** is a diagram of the joint of FIG. **1**, and a spacer of the disclosure;
[0044] FIG. **29** depicts the spacer and joint of FIG. **28**, the joint having been aligned by the spacer;
[0045] FIG. **30** is a diagram of the joint and spacer of FIG. **28**, and a cross-section of a plate of the disclosure, the spacer and plate cooperative to align the joint;
[0046] FIG. **31** depicts a joint of the body, in this example the L5 and S1 vertebrae, and a cross-section of a plate of the disclosure connected to both vertebrae;
[0047] FIG. **32** depicts the joint of FIG. **31**, aligned by the plate, the plate further secured to the joint;
[0048] FIG. **33** is a diagram of the joint of FIG. **31**, and a cross-section of a plate and fastener of the disclosure;
[0049] FIG. **34** depicts the joint and device of FIG. **33**, the joint having been aligned by the device, the fastener shortened after use;
[0050] FIG. **35** is a diagram of the joint of FIG. **1**, and a fastener of the disclosure;
[0051] FIG. **36** depicts the joint and fastener of FIG. **35**, the joint having been aligned by the fastener;
[0052] FIG. **37** is a diagram of the joint of FIG. **1**, and a ratcheting device of the disclosure connected to each side of the joint;
[0053] FIG. **38** depicts the joint and device of FIG. **37**, the joint having been aligned by the device;
[0054] FIG. **39** is a diagram of the joint of FIG. **1**, and a cross-section of a segmented spacer of the disclosure;
[0055] FIG. **40** depicts the joint and spacer of FIG. **39**, the joint secured in alignment, and distracted, using the device of FIG. **39**;
[0056] FIG. **41** is a diagram of the joint of FIG. **1**, and a cross-section of a segmented spacer having three mateable portions, including opposed segments and a wedge shaped spacer;
[0057] FIG. **42** depicts the joint and device of FIG. **41**, the spacer having been inserted between the opposed segments, and the joint, opposed segments, and spacer maintained in alignment by fasteners and a mating engagement between the opposed segments and the spacer;
[0058] FIG. **43** depicts the joint and device of FIG. **40**, the opposed segments shaped to reduce an angle formed between bones of the joint;
[0059] FIG. **44** depicts the joint and device of FIG. **40**, the opposed segments shaped to increase an angle formed between bones of the joint;
[0060] FIG. **45** depicts a diagram of the joint of FIG. **1**, and an expandable spacer, plate, and reaching tool of the disclosure;
[0061] FIG. **46** depicts the spacer and device of FIG. **45** in an expanded configuration, distracting bones of the joint, the joint having been aligned by the spacer, plate, and reaching tool;
[0062] FIG. **47** is a diagram of the joint of FIG. **1**, and a cross-section of a spacer, angled plate, and

fastener of the disclosure;
[0063] FIG. **48** depicts the joint and device of FIG. **47**, the joint having been aligned using the device, the fastener having been shortened after use;
[0064] FIG. **49** is a diagram of the joint of FIG. **1**, and a cross-section of spacer, integrated plate, and fastener of the disclosure;
[0065] FIG. **50** depicts the joint and device of FIG. **49**, the joint having been aligned by the device, the fastener shortened after use;
[0066] FIG. **51** is a diagram of the joint of FIG. **1**, a cross-section of a spacer of the disclosure with ramped segments, and an indication of an application of force to align the joint;
[0067] FIG. **52** depicts the joint and spacer of FIG. **51**, the joint having been aligned using the spacer, the spacer segments fastened together after alignment;
[0068] FIG. **53** is a diagram of the joint of FIG. **1**, and a cross-section of an adjustable segmented spacer of the disclosure, the spacer including a rotatable screw operative to align the segments;
[0069] FIG. **54** depicts the joint and spacer of FIG. **53**, the joint having been aligned using the spacer, the segments of the spacer fixed by a fastener;
[0070] FIG. **55** is a side view of a combined plate, spacer, and fasteners of the disclosure;
[0071] FIG. **56** is a perspective view from the top of the device of FIG. **55**;
[0072] FIG. **57** is a side view of an alternative combined plate, spacer and fasteners of the disclosure;
[0073] FIG. **58** is top perspective view from the top of the device of FIG. **57**;
[0074] FIG. **59** is an alternate top perspective view from the top of the device of FIG. **57**;
[0075] FIG. **60** is a side view of a mateable plate and spacer of the disclosure;
[0076] FIG. **61** is a perspective view of the device of FIG. **60**;
[0077] FIG. **62** is a side view of the device of FIG. **60**, the plate and spacer mutually assembled in conforming relationship;
[0078] FIG. **63** is a front perspective view of a combined plate and spacer of the disclosure;
[0079] FIG. **64** is a front perspective view of a combined plate and spacer of the disclosure, the plate provided with a shaped projection;
[0080] FIG. **65** is a perspective illustration of spacer of the disclosure formed of mateable segments;
[0081] FIG. **66** is a perspective view of the spacer of FIG. **65**, the segments mated;
[0082] FIG. **67** is a top view of the device of FIG. **65**;
[0083] FIG. **68** is a top view of the device of FIG. **66**;
[0084] FIG. **69** is a back view of a combined spacer and plate in two segments, the segments slidingly engageable along an A-P axis;
[0085] FIG. **70** is a side view of the device of FIG. **69**, the segments partially slidingly engaged;
[0086] FIG. **71** is a perspective view of the device of FIG. **70**;
[0087] FIG. **72** is a side view of a combined spacer and plate in two segments, the segments slidingly engageable along a lateral axis;
[0088] FIG. **73** is a perspective view of the device of FIG. **72**;
[0089] FIG. **74** is a side view of a combinable spacer, plate, and fasteners of the disclosure, the plate provided with a shaped projection, the fasteners disposed at an angle;
[0090] FIG. **75** is a side view of a combinable spacer, plate, and fasteners of the disclosure, the plate provided without the shaped projection illustrated in FIG. **74**, the fasteners disposed at an angle;
[0091] FIG. **76** is a bottom perspective view of a combinable spacer, plate, and fasteners of the disclosure, the plate provided with a shaped projection, the fasteners disposed at a converging angle;
[0092] FIG. **77** is a top perspective view of the device of FIG. **76**;
[0093] FIG. **78** is a front view of the device of FIG. **74**, illustrating locking pins;

[0094] FIG. **79** is a side view of the device of FIG. **74**, implanted within a spine;
[0095] FIG. **80** is an illustration of an insertion and alignment tool in accordance with the disclosure;
[0096] FIG. **81** depicts a side view of the tool of FIG. **80** gripping the device of FIG. **74**;
[0097] FIG. **82** depicts the tool and device of FIG. **81**, the tool having inserted the device into a spine, the device fastened to the S1 vertebra;
[0098] FIG. **83** depicts the tool and device of FIG. **81** having aligned vertebrae of the spine;
[0099] FIG. **84** depicts the tool and device of FIG. **83**, the device fastened to the L5 vertebra;
[0100] FIGS. **85**A-**85**C depict a method of inserting a spacer in accordance with the disclosure;
[0101] FIG. **86** depicts a side view of a combinable spacer, plate and fasteners of the disclosure, the plate provided with an upper lip; and
[0102] FIG. **87** depicts a joint of the body, in this example the L5 and S1 vertebrae, and a cross-section of an alternative plate of the disclosure connected to both vertebrae.

DETAILED DESCRIPTION

[0103] As required, detailed embodiments are disclosed herein; however, it is to be understood that the disclosed embodiments are merely examples and that the systems and methods described below can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present subject matter in virtually any appropriately detailed structure and function. Further, the terms and phrases used herein are not intended to be limiting, but rather, to provide an understandable description of the concepts.

[0104] The terms “a” or “an”, as used herein, are defined as one or more than one. The term plurality, as used herein, is defined as two or more than two. The term another, as used herein, is defined as at least a second or more. The terms “including” and “having,” as used herein, are defined as comprising (i.e., open language).

[0105] With reference to FIG. **1**, a healthy pair of vertebrae **300**, **302** are diagrammatically illustrated. Although adjacent vertebrae are not identical in the body, to simplify an understanding of an operation of devices and methods of the disclosure, a centrally disposed dotted line **304** represents a natural overlapping of bones. The disc annulus surrounding the nucleus pulposus is diagrammatically illustrated as disc **306**, positioned in alignment with both vertebrae. In FIG. **2**, vertebrae **300**, **302** are relatively displaced, representing a patient condition addressed by the disclosure, for example degenerative fibrocartilage, malignancy, herniation, protrusion, spondylolysis, spondylolisthesis, spinal stenosis, or degenerative disc disease. While one vertebra **300** appears to be oriented in an upper, or superior location with respect to the other vertebra **302**, it should be understood that within the body, superior, inferior, or both bones may be displaced, and that a superior or inferior position of adjacent bones should not be assumed within the disclosure. In addition, a displacement may be anterior, posterior, or lateral.

[0106] In the various embodiments described herein, the spine and vertebrae are described as the bones to be aligned. However, it should be understood that the methods and devices of the disclosure are readily applied or adapted to be used with other bones of the body, for example in size and shape. Such other bones include the joints of the fingers, hand, wrist, elbow, shoulder, hip, knee, ankle, foot, or toes. Non-jointed tissue, such as bony plates, for example misaligned after trauma, may also be aligned as described herein. Accordingly, the disclosure should be considered in light of other bones and tissue, throughout.

[0107] In various embodiments, the devices and methods of the disclosure provide solutions for treating diseases of the spine, for example, isthmic spondylolisthesis, from a single approach, such as advantageously an anterior approach. This avoids a requirement of an approach through two separate aspects of the body, for example also a posterior approach. This reduces trauma to the patient, recovery time, time to complete the surgical procedure, risk, and cost. While an anterior approach is possible, it should be understood that a lateral or posterior approach is similarly possible, while retaining the advantage of aligning bones of a joint from a single approach, including only one of a left lateral approach, a right lateral approach, an anterior approach, or a posterior approach.

[0108] In FIG. **3**, an intravertebral spacer **5000** engages one or more fasteners, for example bone screws **200**, for example by passing a polyaxial screw through an aperture, not shown, or through some other configuration understood within the art. In this manner, a position of spacer **5000** may become fixed with respect to one or more vertebrae, at least for a sufficient period of time for healing and or fusion between the body and the spacer to take place, for example through bony ingrowth. Spacer **5000** may be provided with spaces, channels, texture, or other structure which encourages bone ingrowth, and may further be provided with therapeutic substances which encourage bone growth.

[0109] With further reference to FIG. **3**, an assembly in accordance with the disclosure includes a plate **400**, fasteners or screws **200**, **202**, and an intervertebral spacer **5000**. Spacer **5000** is secured to a vertebra **300** using one or more bone screws **200** or other fastener. It should be understood that while a bone screw **200** is illustrated in the drawings, for clarity, any fastener suitable for connecting an implant to a bone may be used, including adhesive, as would be understood by one skilled in the art. In this configuration, vertebra **300** is closer to the surgical approach than vertebra **302**. For example, in an anterior approach, vertebra **300** is shifted anteriorly with respect to vertebra **302**. In a lateral or posterior approach, plate **400** would be applied to the more lateral or posterior vertebra, respectively.

[0110] Plate **400**, in accordance with this and other embodiments of the disclosure, may have a flat, rectangular or bar shape such as is illustrated in the drawings, for clarity, or may be contoured and or curved, or shaped to closely conform, to the body tissue to which it is attached. In some embodiments, the plate **400** is rigid, while in other embodiments, the plate **400** can be flexible or bendable. In some embodiments, the plate **400** is configured such that in an unstressed state, the plate **400** is flat, while in a stressed state, it is curved.

[0111] After plate **400** is secured to vertebra **300**, a screw **202** or other fastener configured to perform a like function, is connected between plate **400** and spacer **5000**. Screw **202** is rotatably connected to plate **400** whereby as screw **202** is rotated, spacer **5000** is drawn closer to plate **400**. Herein, screws should be considered as levers; accordingly, screw **200** functions as a lever to facilitate movement of one vertebra with respect to another. Vertebra **302**, being connected to spacer **5000**, is likewise drawn closer to plate **400** as screw **202** is rotated, the assembly thereof ultimately causing the therapeutically correct disposition of vertebrae **300**, **302** shown in FIG. **4**. As is the case with this and other embodiments of the disclosure, after correction, the assembly may be permanently left in the body, or may be removed after a therapeutic period of time.

[0112] Referring now to FIGS. **5**A, **5**B, and **6**, vertebrae **300** and **302**, as well as associated ligaments **308**, are relatively displaced. In this embodiment, an expanding spacer **5002** is positioned between vertebrae **300**, **302**, for example after removal of all or part of intervertebral tissue, or disc **306**A. In this illustration, an additional vertebral level is shown, including vertebra **300**A. It should be understood in all embodiments herein that more than one level may be corrected using methods and devices of the disclosure. For example, multiple plates may be used, or longer plates spanning more than one level.

[0113] In FIGS. **5**A, **5**B, and **6**, spacer **5002** may be of any known or hereinafter invented type, including spacers having, for example, mechanical or hydraulic mechanisms of expansion. Vertebra **302** is connected to plate **400** with bone screw **200**. In one embodiment, spacer **5002** is connected to plate **400**, to ensure that spacer **5002** remains in position with respect to vertebra **302**, although this is not needed in all cases. For example, spacer **5002** may be separately engaged with vertebra **302**. Tool **6000** includes a brace **6002** and a driver **6004** moveably connected to brace **6000**. Brace **6000** is releasably engageable with plate **400** at connector **6006**.

[0114] To correct an alignment of the spine, driver **6004** is moved with respect to brace **6002**, for example through a threaded, hydraulic, motor, or other moveable attachment between driver **6004** and brace **6002**. It should be understood that herein, drawings are not necessarily to scale, and that portions of spacers, plates, their assemblies, or insertion and reduction tools may be sized to be inserted in their entirety, or as portions to be assembled, through a minimally invasive incision or through one or more percutaneous punctures, for example using an endoscope or cannula. As brace **6002** is connected with plate **400**, a movement in the direction of arrow “A” in FIG. **5**A causes a displacement in the direction “A” of vertebra **300** and associated ligaments **308**. Prior to, during, or after movement of vertebra **300**, spacer **5002** may be expanded to dispose vertebrae **300**, **302** at a correct relative distance, and to provide further stability to the spine. In FIG. **6**, tool **6000** is detached, and if needed, an additional bone screw **200** may connect plate **400** and vertebra **300**. One or more screws **200** may alternatively pass through plate **400**, spacer **5002**, and or vertebra **300**. In an alternative embodiment, plate **400** is not used, and brace **6002** is releasably connectable to expanding spacer **5002**.
[0115] With reference to FIGS. **7-9**, an assembly of the disclosure includes a tool **6010** cooperative with plate **400**, attached to vertebra **300** by screws **200**. Tool **6010** includes a chassis **6012** extendable around a vertebra, and moveably attached to plate **400**. In some embodiments, the chassis **6012** can be formed of multiple pieces, or of a single piece. More particularly, plate **400** attaches to a first vertebra **300**, extends proximate a second vertebra **302** which it is desired to reduce into alignment with the first vertebra **300**. Chassis **6012** includes one or more extensions attached to plate **400**, extending to engage vertebra **302** about a side of vertebra **302** opposite to a location of plate **400**, to thereby at least partially encircle vertebra **302** to effect a positive engagement. It should be understood, however, that chassis **6012** may frictionally engage vertebra **302** along a side surface of vertebra **302**, or may be connected to vertebra **302** with one or more fasteners.
[0116] In the embodiment shown, screw **202** threadably engages chassis **6012** and rotatably contacts plate **400**, whereby as screw **202** is turned, chassis **6012** is pushed away from plate **400**, thereby drawing vertebra **302** into alignment with vertebra **300**. When a desired alignment is accomplished, vertebra **302** may be attached to plate **400** to maintain the alignment, or vertebra **302** may be fastened to spacer **5000**, or to vertebra **300**, by any other known or hereinafter invented means, if needed. Once vertebra **302** is sufficiently stable in its new disposition, tool **6010** may be removed.
[0117] In FIGS. **10** and **11**, an assembly of the disclosure includes a spacer **5010** which is formed of two mating portions **5012**, **5014**, and forms, in the illustrated embodiment, an angular external profile corresponding to a desired disposition of adjacent vertebrae **300**, **302**. It should be understood that the illustrated concept applies equally to non-angular or other external profiles. Mating portions **5012**, **5014** are provided with serrations, teeth, or other interlocking engagements **5016**, **5018**, whereby when mating portions **5012**, **5014** are overlapped completely or to a desired extent, a relative position of mating portions **5012**, **5014** is fostered, particularly when mating portions **5012**, **5014** are pressed together, for example between vertebrae **300**, **302**, by natural compression and associated ligaments **308** (not shown). Mating portions **5012**, **5014** may also be secured together by a fastener, including a screw or adhesive.
[0118] In FIGS. **10-11**, interlocking engagements **5016**, **5018** have the form of ramped teeth, whereby a mutual sliding engagement, advantageously in a direction of therapeutic alignment of vertebrae **300**, **302**, is facilitated, but discouraged in a direction away from therapeutic alignment, as may be seen in FIGS. **10-11**. A tool **6018**, diagrammatically illustrated in FIG. **10**, urges, translates, or pushes the vertebra **300** in a desired direction indicated by arrow “A”. Tool **6018** may be a mechanical device, for example as described within this disclosure, or may be a medical practitioner's hand.
[0119] In an embodiment illustrated in FIGS. **12-13**, mating portions **5012**, **5014** may or may not

have interlocking engagements, as shown, but are secured in mutual engagement by a fastener, such as screw **202**, which is advantageously the same screw as used to manipulate mating segment **5014** in FIG. **10**. Screw **202** may advantageously be passed through a first vertebra **300** and threadably fastened into a second vertebra **302**.

[0120] Tool **6018**, diagrammatically illustrated in FIG. **14**, may be configured and or used to engage both spacer **5000** and vertebra **300**, advancing both together until a therapeutic alignment is attained. After reduction, vertebrae **300**, **302** and spacer **5000** may be secured as described elsewhere within the disclosure.

[0121] In FIG. **15**, vertebra **300** is reduced in sliding engagement with spacer **5000**. After reduction, natural or induced compression, for example by compression due to body weight, or engagement with bone screws, promotes insertion of textured surface or projections **5020** with bone or body tissue, securing a position of spacer **5000** and relative alignment of vertebrae **300**, **302**.

[0122] In FIG. **16**, an assembly of the disclosure includes instrument or tool **6020**, used to reduce vertebra **300** in cooperation with spacer **5030**. A brace **6022** abuts or connects with spacer **5030**, which may be adapted to securely engage brace **6022** during reduction. In the example of FIGS. **16-18**, a threaded connection **6024** is used, although other types of releasable mechanical abutments or interlocks may be used. Once brace **6022** is connected with spacer **5030**, it may serve to align and guide a push rod **6026**, slideably connected to brace **6022** by a guide **6028**, in this embodiment a guide ring attached to brace **6022** through which push rod **6026** slideably passes. An engagement surface **6030** is sized and shaped to move within the body, securely contact body tissue, and urge bones into alignment when push rod **6026** is pushed towards body tissue. Once vertebrae **300**, **302** are aligned, tool **6020** may be disconnected from spacer **5030** and removed from the body. In one embodiment, tool **6020** is sized and dimensioned to be used through a minimally invasive percutaneous opening in the skin, or using endoscopic techniques.

[0123] As with the embodiment of FIGS. **16-18**, the assembly of FIG. **19** engages a tool to a spacer, to anchor and orient a tool for displacing and realigning a bone. In the embodiment of FIG. **19**, tool **6032** is releasably connected to spacer **5000**, in this embodiment with screw **202**, and includes a lever **6034** which transmits a displacement force to an adjacent vertebra **302**. A threaded adjuster **6036** includes a screw **6038** rotatably mounted to a guide **6040**, screw **6038** operative to advance an engagement surface **6042** to contact and move vertebra **300**. As resistance is imparted to tool **6032**, force is transferred to screw **202**, and a rotational moment of tool **6032** is transferred to lever **6034**. In this manner, the vertebrae **300**, **302** to be relatively aligned share a load force, and are each displaced to an extent, in opposite directions. This is further advantageous at least for the reasons that the tool is stabilized, and a displacement force imparted by tool **6032** is shared between spacer **5000** and vertebra **302**. In some embodiments, a biasing member, such as for example an optional spring **6044**, may advantageously be included to bias engagement surface **6042** in a direction of vertebra **300**. Additionally or alternatively, the spring **6044** may limit an extent of force imparted to vertebra **300** by screw **202** and engagement surface **6042**, by enabling screw **6038** to move in an opposite direction to an applied force against vertebra **300**. In other embodiments, the screw **6038** acts on its own without the assistance of a biasing member, such as spring **6044**.

[0124] With reference to FIG. **20**, an assembly of the disclosure includes a plate **410** which has one or more guide extensions **412** operative to maintain a relative alignment of plate **410** and vertebra **302** as vertebra **300** is reduced. More particularly, one or more screws **412** may be used to draw vertebra **302** towards a conforming engagement with plate **412**. Concomitantly, an upper portion **414** of plate **410** contacts vertebra **300**, which is displayed in a manner such as is illustrated in FIG. **19**. When screws **412** are fully seated, plate **410** is in conforming engagement with vertebra **300** and **302**, upper portion **414** having urged vertebra **300** into alignment with vertebra **302**, as shown. Guide extensions **412**, extending for example on opposing sides of vertebra **302**, may be secured to vertebra **302** to maintain a lateral alignment of plate **410** with respect to vertebra **302**. Upper portion **414** may be secured to vertebra **300**, or as shown, may moveably contact vertebra **300**, allowing an extent of natural movement between vertebrae **300**, **302**.

[0125] In FIG. **21**, an assembly of the disclosure includes tool **6050** having a lever **6052** pivotally connected to plate **418** at hinge **6054**. Plate, **418** is connected to vertebra **302** by a fastener, such as one or more screws **200**. A lever screw **6056** is moveably connected to lever **6052**, and is connected to vertebra **300**. As lever **6052** is pivoted, for example by application of a manual or mechanical force, vertebra **300** is moved correspondingly. When a desired alignment of vertebra **300** is attained, tool **6050** and plate **418** may be removed. Alternatively, hinge **6054** may be disassembled, and plate **418** may remain in the body. In such an embodiment, plate **418** may be configured to fuse vertebrae **300**, **302**, for example by being taller than illustrated, wherein screw **200** may be passed through plate **418** into vertebra **300**. In another alternative, screw **6056** may be seated to secure lever **6052** against vertebra **300**. An extending portion **6058** of lever **6052**, if present, may be removed, for example by disassembly, or by being separated from a remainder of lever **6052** at a weakened portion **6060**. In this configuration, articulation is preserved or provided between vertebrae **300** and **302**, at hinge **6054**.

[0126] With reference to FIGS. **22-23**, an assembly of the disclosure includes a lever **6064** which engages lever points **5046**, **5048**, shown as projecting pins on separate portions **5042**, **5044** of spacer **5040**. Lever **6064** is advantageously sized and shaped to lever vertebrae **300**, **302** into a correct relative alignment by applying an opposing force to lever points **5046**, **5048**. After a therapeutic alignment has been carried out, portions **5042**, **5044** of spacer **5040** may be affixed relative to each other by a fastener, for example screw **202**, as shown in FIG. **23**. Projections **5046**, **5048** may be removed after use. Alternatively, it should be understood that lever points **5046**, **5048** may be recesses or cutouts formed in portions **5042**, **5044**, thereby reducing an overall profile of spacer **5040**.

[0127] FIGS. **24** and **25** are analogous to FIGS. **22-23**, however lever **6070** engages one of separate portions **5042**A, **5044**A, of spacer **5040**A, and body tissue associated with an opposed vertebra. Lever **6070** thereby directly pushes body tissue of vertebra **300**, obtaining leverage provided by lever point **5048**, attached to an adjacent vertebra **302**. FIGS. **22-23** and **24-25** illustrate differing forms of spacer, **5040** and **5040**A, in order to demonstrate additional advantageous embodiments of spacer. It should be understood that spacers illustrated throughout the specification may be substituted with each other, as would be understood by one skilled in the art.

[0128] In FIGS. **26-27**, an assembly of the disclosure includes plate **420** which is fastened to vertebra **300**, and a screw **200** is rotatably connected to plate **420** and threadably engaged with vertebra **302**, whereby progressive rotation of screw **200** draws vertebrae **300**, **302** into mutual conforming contact with plate **420**. As with all embodiments herein, screw **200** may be any form of bone screw best adapted to the body tissue of the patient and the forces of the application, including for example a Gruberger or lag screw. One or more washers **220**, for example polymeric or metallic washers, may additionally be used to avoid damage to an implanted component, or to avoid the production of debris. After the bones are reduced, plate **420** may be removed, as illustrated in FIG. **27**, or may remain in the body for a period of time, for example until vertebra **300**, **302** are stable in relative relation, or have fused if fusion is desired. In FIG. **27**, fasteners such as screws **200** are illustrated, maintaining vertebrae **300**, **302** in relative position through engagement with spacer **5000**. In the various embodiments, spacer **5000** is illustrated for simplicity and clarity. It should be understood, however, that a position of vertebrae **300**, **302** may be maintained by any other known or hereinafter developed method, including for example rods, dynamic fixation, articulating plates, or cables. In embodiments, lever **6064** or lever **6070** has the form of an elongated bar.

[0129] FIGS. **28-29** illustrate a spacer **5050** having non-parallel bone engaging surfaces **5052**, **5054** forming a wedge shape. Any of the methods and apparatus of the disclosure may be used to align vertebrae **300**, **302**, together with a wedge shaped spacer, such as spacer **5050**. FIGS. **28-29** further illustrate that a therapeutic alignment is any alignment which is beneficial for the patient, whether

natural and expected, or unique to an individual patient.
[0130] FIG. **30** illustrates reduction of bones **300**, **302**, with an assembly of the disclosure which includes a wedge shaped spacer **5050**, and which uses the apparatus and method described above with respect to FIGS. **28-29**, although other methods and apparatus as described herein may be used as would be understood by a medical practitioner.
[0131] FIGS. **31-32** illustrate an apparatus and method similar to that of FIGS. **26-27** as employed at the L5-S1 level. While the sacrum **310**, in particular, is illustrated here, it should be understood that vertebra **300**, **302**, **310** may be any vertebrae or jointed surface of the body.
[0132] FIGS. **33-34** also illustrate an apparatus and method similar to that of FIG. **26**, however in this embodiment, screw **204** is provided with threads **206** which engage a nut **208** at one end, and body tissue at an opposite end. Screw **204** is passed through, or otherwise engages plate **420**, for example with a notch (not shown), and is fastened to body tissue. Two nuts secured together may be employed at the end of screw **204** for engaging a driving tool, or alternatively, screw **204** may be provided with a tool engaging surface, for example a hex, slotted, or Phillips engagement (not shown). If such engagement interferes with passage of nut **208** onto screw **204**, nut **208** may be passed first along the bone engaging end of screw **204**. After screw **204** is engaged with body tissue, in this example sacrum **310**, and plate **420**, nut **208** may be driven upon screw **204** and in engagement with plate **420**, to move plate **420** and sacrum **310** together, either by displacing sacrum **310**, vertebra **300**, or a combination of both, until alignment is achieved, as shown in FIG. **34**. After alignment, screw **204** may be reduced in length to avoid interference with body tissue, particularly after an opening in the body is closed. While screw **204** may be severable with a tool, for example a saw or cutting shears, one or more weakened regions **210** may be provided along the length of screw **204** to facilitate breaking off or otherwise removing an excess length of screw **204**.
[0133] FIGS. **35-36** illustrate reduction of vertebrae **300**, **302** using an elongate screw **212** which passes through the intervertebral space, connecting two adjacent bones. Threads **214** are formed on a distal end of screw **212**, whereby after passing through and seating within vertebra **300**, screw **212** may thereafter freely rotate in connection with vertebra **300**. Screw head **216** is sized to block movement of screw through the cortical bone of vertebra **300**. Once screw **212** is seated within vertebra **300**, threads **214** continue to engage tissue of vertebra **302**, thereby drawing vertebra **302** towards vertebra **300**, and into a therapeutic alignment. After alignment, screw **212** may be left within the body, or removed, optionally replaced by alternative means for maintaining an alignment of vertebrae **300**, **302**.
[0134] With reference to FIGS. **37-38**, assembly **128** includes a ratchet tool **6080** having a lever **6082** attached to a toothed rack, pawl, or gear **6084**, engaged with one or more mating racks **6086**. A drawbar **6088** is releasably engaged with vertebra **300**, for example with angled teeth **6090**. More particularly, teeth **6090** may be slidingly positionable with respect to vertebra **300** in a first direction, and engageable with tissue associated with vertebra **300** in a second direction, whereby vertebra **300** may be reduced in the second direction when drawbar **6088** is engaged and displaced. Drawbar **6088** is connected to rack **6086**, whereby movement of rack **6086** causes a corresponding movement of drawbar **6088**, whereby a medical practitioner may move an engaged vertebra in a second direction. In one embodiment, gear **6084** is rotatable about a fixed point **6092** with respect to vertebra **300**, whereby movement of lever **6082** causes reduction of vertebra **300** with respect to fixed point **6092**.
[0135] In another embodiment, a second drawbar **6088**A engages a second vertebra **302**, however in this embodiment angled teeth **6090**A slidingly engage vertebra **302** in the second direction, and engage tissue associated with vertebra **302** in the first direction. Rack **6086**A additionally engages gear **6084**, whereby movement of lever **6082**, causes movement of drawbars **6088**, **6088**A in opposite directions, thereby applying a moving force to both vertebrae **300**, **302**, but in opposing directions, promoting relative alignment of vertebrae **300**, **302**.
[0136] In FIGS. **39-40**, an assembly of the disclosure includes a spacer **5058** having two mating

portions **5060**, **5062**, each releasably or fixedly engageable with a vertebra at a bone engaging surface **5064**, **5066**. Mating surfaces **5070**, **5072** are connected to move relative to each other, joined by a sliding fastener **5074**. An adjusting device **5076** is associated with sliding fastener **5074** to retain a relative position of mating portions **5060**, **5062**, together with engaged vertebra **300**, **302**, once sliding alignment is carried out. In the embodiment illustrated, adjusting device **5076** includes a threaded rod **5078** engageable within a channel **5080**, **5082** disposed in each of mating surfaces **5070**, **5072**, however it should be understood that other mechanical couplings are contemplated within the disclosure, as would be understood by one skilled in the art, for example including a riveted or molded coupling. Advantageously, threaded rod **5078** may be threadingly received within mating surfaces **5070**, **5072**, whereby a distance between vertebrae **300**, **302**, for example distraction, may also be carried out, in addition to reduction. Additional threaded nuts, or other fastener, not shown, may be used along threaded rod **5078** to secure adjusting device **5076** at a desired location along channel **5080**, **5082**.

[0137] Referring now to FIGS. **41-44**, a wedge shaped spacer **5090** includes recessed or projecting engaging surfaces **5092**. Spacer segments **5096**, **5098** are sized and dimensioned to be connectible with facing portions of vertebrae **300**, **302**, and are provided with projecting or recessed engaging surfaces **5100** complementary to and mateable with engaging surfaces **5092**. As spacer **5090** is inserted between vertebra **300**, **302**, a distraction of vertebra **300**, **302** takes place due to the wedge shape, and the force with which spacer **5090** is inserted. When spacer **5090** is sufficiently inserted, engaging surfaces **5092** and **5100** will engage, initially in connection with one of segments **5096**, **5098**. Subsequently, the medical practitioner displaces or reduces one or both of vertebrae **300**, **302** until they are therapeutically aligned. Manipulation of vertebrae **300**, **302** may be carried out by engaging segments **5096**, **5098** manually, or with a tool, for example using screws **218**, or other engageable surface. Alternatively, or additionally, vertebrae **300**, **302** may be directly manipulated.

[0138] Engaging surfaces **5092** and **5100** are positioned so that a complementary engagement takes place when a therapeutic alignment of segments **5096**, **5098**, and spacer **5090** has been accomplished. Accordingly, after alignment, an engagement is made between engaging surfaces **5100** of segment **5098** and engaging surfaces **5092** of spacer **5090**, and engaging surfaces **5100** of segment **5096** and spacer **5090**, thereby aligning all of vertebra **300**, segment **5096**, spacer **5090**, segment **5098**, and vertebra **302**.

[0139] After alignment, body tissue associated with vertebrae **300**, **302**, such as ligaments **308**, maintain engagement of segments **5096**, **5098**, and spacer **5090**. Additionally, spacer **5090** may be further secured to either or both of segments **5096**, **5098** by fasteners, including adhesives or threaded fasteners. Similarly, segments **5096**, **5098** may be secured to body tissue.

[0140] FIGS. **43-44** illustrate alternative configurations of the embodiment of FIGS. **41-42**, in which segment **5096**A or **5098**A have a sloped shape, adapted to complement a wedge shape of spacer **5090**. In this embodiment, a relative angular alignment of vertebrae **300**, **302** is not changed, however an anterior-posterior displacement may be corrected. Alternatively, an increased angular alignment may be achieved by the use of an alternative angular profile of segment **5096**B and or segment **5098**B, as shown in FIG. **44**.

[0141] With reference to FIGS. **45-46**, any tool of the disclosure may be used to address a displacement of vertebrae **300**, **302**, to form an assembly including a dynamic spacer **5110** which is used to distract a joint formed between vertebrae **300**, **302**. In this embodiment, exemplary tool **6010**, as described with respect to FIGS. **8-10**, is used to reduce vertebrae **300**, **302**. During or after the reduction of a displacement of vertebrae **300**, **302**, distraction is achieved using spacer **5110**, which may be resilient, or which may be expanded by the introduction of a gas or fluid, for example purified air or sterile saline. After distraction, any known form of stabilization may be employed, for example including a plate, rod, or cable. Spacer **5110**, as with any spacer of the disclosure, may be provided with a surface texture and or a releasable therapeutic substance which, for example, promotes bone growth.

[0142] In FIGS. **47-48**, an assembly of the disclosure is illustrated which is similar to the assembly and method described with respect to FIGS. **33-34**, but is configured to cooperate with spacer **5120**. More particularly, spacer **5120**, which may have the form of any spacer of the disclosure, is secured to plate **420** by any known means, including screw **202**, as illustrated. This further stabilizes vertebrae **300**, **302**, and maintains a therapeutic location of a spacer. In a further embodiment, a connection between spacer **5120** and plate **420** is partly or substantially rigid, preserving at least a portion of a natural range of motion.

[0143] The assembly of FIGS. **49-50** is analogous to plate **420** and screw **204**, and functions in a like manner. However, plate **420** is connected to, or formed together with, spacer **5124**. In this manner, a superior-inferior disposition of vertebrae **300**, **302** may be achieved during insertion of spacer **5124**, followed by a reduction in an anterior-posterior direction during tightening of nut **208**. This embodiment further provides for fixation of adjacent vertebrae. Either or both of extensions **5126**, **5128** may be flexible, to preserve at least a portion of a natural range of motion.

[0144] In FIGS. **51-52**, a spacer **5130** includes two segments **5132**, **5134**, each connectable to a vertebra **300** or **302**. Each segment **5132**, **5134** has a ramped surface **5136**, **5138** complementary to the other, whereby as segments **5132**, **5134** are moved relative to each other into an overlapped conformity, a height of spacer **5130** is increased. In this manner, segments **5132**, **5134** may be assembled into an intervertebral space, engaged with body tissue of adjacent vertebrae, and then slide along ramped surfaces **5136**, **5138** into overlapping conformity, whereby reduction and distraction can both be accomplished simultaneously. A sufficient engagement may be achieved between segments **5132**, **5134** and the respective vertebrae **300**, **302** to which they are attached, by projections **5140**, or adhesive. An engagement may be further secured by the use of one or more bone screws **200**.

[0145] In the embodiment of FIGS. **51-52**, reduction may be carried out by apparatus and methods of the disclosure. In one embodiment, a fastener, for example screw **202** in FIG. **51**, may be used to attach to a tool. In FIG. **52**, vertebrae **300**, **302** are aligned, and segments **5132**, **5134** are secured to each other, for example using adhesive or other fastener, such as screw **202**, illustrated. In another embodiment, screw **202** may freely rotate within a slot (not shown) in segment **5134**, and be threadably received within segment **5132**, whereby reduction and distraction may be carried out, at least in part, by rotating screw **202**.

[0146] Referring to FIGS. **53-54**, an embodiment includes a spacer **5150** having segments **5152**, **5154**, each connectable to a vertebra **300** or **302**. A threaded shaft **5156** is disposed between segments **5152** and **5154**, threadably engaging threaded surface **5158** of segment **5152**, and slidingly engaging surface **5160** of segment **5154**. As shaft **5156** is rotated, shaft **5156** engages threaded surface **5158**, driving segment **5152** into alignment with segment **5154**, thereby aligning the attached vertebrae **300**, **302**. One or more shaft collars **5162** prevents movement of shaft **5156** out of engagement with segment **5154**. An alignment of segments **5152** and **5154** may be maintained with a fastener, for example screw **202**, thereby maintaining an alignment of vertebrae **300**, **302**.

[0147] In some embodiments, the threaded shaft **5162** can be configured to allow for a movable upper member (e.g., such as a movable segment **5152**) to move relative to a stationary lower member (e.g., such as a stationary segment **5154**). The movable upper member can be attached to a superior vertebral body, while the stationary lower member can be attached to an inferior vertebral body.

[0148] In FIGS. **55-56**, a plate **430** may be therapeutically used alone, or in combination with other devices of the disclosure, to provide stability to adjacent bones, for example vertebrae **300**, **302**. Plate **430** includes an exterior plate **432**, and a plate extension **434** sized and dimensioned to project into an articulating region of the joint when the exterior plate is fastened to adjacent bones. Plate extension **434** further includes one or more projections **436** configured to pierce body tissue of adjacent bones within the articulating space, thereby securing a location of the adjacent bones, plate extension **434**, and exterior plate **432**, relative to each other. In this manner, after adjacent bones, for example bones **300**, **302** are aligned in accordance with the disclosure, plate **430** may be inserted and secured within the intervertebral space to prevent, through engagement of projections **436**, to prevent a migration or movement of vertebrae **300**, **302** out of alignment. To further secure vertebrae **300**, **302** and plate **430** relative to each other, bone screws **200** may be passed through apertures **438**. A locking screw or pin **442** (an example is shown in FIG. **78**) may be attached at lock location **440**, to prevent reversal of screws **200** after installation.

[0149] In FIGS. **57-59**, spacer **5170** is connectable to plate **444** by one or more fasteners, for example one or more screws **202** (not shown). One or more bone screws **200**, or other fastening method of the disclosure, may be used to secure spacer **5170** in a joint space, for example in the intervertebral space between vertebrae **300**, **302**, prior to connecting plate **444**. In this manner, additional space is provided for accessing the joint space. Plate **444** may thereafter be connected to spacer **5170**. As an alternative to connecting plate **444** to spacer **5170** directly, one or more bone screws **200** may be passed through apertures **446** in plate **444**, through apertures **5176** in spacer **5170**, and into body tissue. After connection to spacer **5170**, plate **444** may be connected to opposing sides of the joint, for example to cortical bone of vertebrae **300**, **302**. In this manner, an alignment of vertebrae **300**, **302** may be preserved, as vertebrae **300**, **302** are connected to each other at least through a mutual connection to plate **444**, and additionally via spacer **5170**, either through screws **200**, and or via projections **5172**, or by bone ingrowth, for example through an aperture **5174** in spacer **5170**. Finally, additional support is achieved by connecting spacer **5170** and plate **444**, as described.

[0150] In FIG. **60**, spacer **5170** is utilized, as described with respect to FIGS. **57-59**, however plate **448** is provided with one or more projections or posts **450** which mateably engage spacer **5170**. In this manner, spacer **5170** may be inserted to a desired depth within the joint space, and post **450** may be engaged with spacer **5170** to prevent a change in angular disposition between plate **448** and spacer **5170**. In the embodiment shown, plate **448** is further connectable with one of the bones of the joint, for example vertebra **300**, however plate **448** could extend in an opposite direction to be more easily connectable with a bone on the opposite side of the joint, for example vertebra **302**. One or more bone screw apertures **446** and lock locations **440** may be provided as described herein. A fastener, for example screw **202**, may be used to further engage plate **444** and spacer **5170**, and to prevent movement of spacer **5170** within the joint space. Tool engagement surfaces **452**, **5178** upon plate **448** and spacer **5170**, respectively, are configured to facilitate releasable attachment of a surgical tool to an implanted device of the disclosure, for case of manipulation.

[0151] In another embodiment of the disclosure, shown in FIG. **63**, plate **456** connects to spacer **5180** by a fastener as disclosed herein, or by a dovetail or other shaped connection. Spacer **5180** is not provided with apertures **5176** (as shown for example in FIGS. **58** and **62**), or if they are present, they are not required to be used. More particularly, plate **456** provides for anchoring both plate **456** and spacer **5180** to the joint, using angulated apertures **458**, through which bone screws or other fastener may be passed to connect with bones of the joint, for example vertebrae **300**, **302**. Reduction of vertebrae **300**, **302** may be accomplished as disclosed elsewhere herein, or plate **456** may be fastened to first one vertebra **300**, and then as bone screws are subsequently rotated into bone of vertebra **302**, plate **456** and vertebrae **302** are drawn together, positioning vertebrae **300**, **302** into alignment.

[0152] The embodiment of FIG. **64** is similar to the embodiment of FIG. **63**; however plate **460** includes one or more shaped projection **464** that function to conform to the particular shape of a bone to which it is engaged, such as the sacrum, to form a better, more stable fit, and to produce a smaller implanted profile. Locking locations **440** are advantageously provided to prevent reversal of a fastener inserted through bone screw apertures **446**.

[0153] In FIGS. **65-68**, spacer **5200** includes two mateable segments **5202**, **5204** which fasten together using a ratchet interface **5206**. Segment **5202** forms an intervertebral spacer portion **5208** advantageously including projections **5210** which engage body tissue in the joint space to reduce a likelihood of movement of bones of the joint relative to segment **5202**. One or more mounting extensions **5212** extend from intervertebral spacer portion **5208** and are configured, for example with one or more apertures **5236**, to engage a fastener to secure segment **5202** to body tissue, for example by passage of a bone screw **200** through mounting extension **5212** and into cortical bone of one of vertebrae **300**, **302**.

[0154] Segment **5204** slideably engages segment **5202**, formed to project into and nest within a groove **5214** on an inner face **5216** of intervertebral spacer portion **5208**. Adjacent to, or incorporated within groove **5214** and mating portions of segment **5204**, are ramped engagement surfaces **5218**, **5220** which admit passage past each other as segment **5204** is slidingly engaged with segment **5202** to pass into an interior passage **5222** of segment **5202**, and which prevent a reversal of this engagement of segments **5202**, **5204**. Engagement surfaces **5218**, **5220** form an interference fit with respect to each other, and are permitted to engaged successive mutually facing ramped surfaces by a resilient bending of the material of segments **5202**, **5204**. It should be understood that groove **5214** may be formed within segment **5204**, and segment **5202** could be formed to extend thereinto.

[0155] Segment **5204** is similarly provided with an interior passage **5224**, whereby the mated segments form an opening between vertebrae **300**, **302**, through which bone growth may therapeutically take place, for further stabilization, and to promote tissue health. Longitudinal projections **5228** may advantageously be formed upon an upper and lower face of an intervertebral spacer portion **5230**, aligned along the direction of mateable sliding of segment **5204** with segment **5202**, and similarly serve to provide a stable engagement between segment **5204** and body tissue into which projections **5228** project.

[0156] Segment **5204** further has one or more mounting extensions **5226** extending from an intervertebral spacer portion **5230**. As with mounting extensions **5212**, mounting extensions **5226** are configured to engage a fastener to secure segment **5202** to body tissue, for example by passage of a bone screw **200** through mounting extension **5212** and into cortical bone of one of vertebrae **300**, **302**.

[0157] To correct for a misalignment of bones of the joint, segment **5202** is first securely connected to a bone of the joint, for example vertebra **300**, using mounting extensions **5212**. Next, segment **5204** is mounted to segment **5202** by sliding segment **5204** partially along a length of groove **5214**, until mounting extension **5226** contacts tissue of the misaligned bone, for example vertebra **302**. Next, using any of the methods described herein, including manually pushing vertebra **302**, segment **5204** is driven further into mateable engagement with segment **5202**, whereby engagement surfaces **5218**, **5220** cooperate to prevent reversal of the engagement, and maintain segments **5202**, **5204** at a desired extend of mutually overlapped engagement. Through engagement with segments **5202**, **5204**, bones of the joint are maintained in relative alignment. Thus secured, no additional plates or stabilizing implants are required, although they may be used as deemed therapeutically beneficial for the patient.

[0158] Segments **5202**, **5204** may be moved relative to each other by attaching a lever between segments **5202**, **5204**, using a method or device as described herein.

[0159] A fastener, including for example a screw or adhesive, may be used to further secure segments **5202**, **5204** in mutual conformity, or to secure segments **5202**, **5204** to body tissue. Bone growth materials, therapeutic drugs, or other beneficial substances may be placed within interior passage **5224**.

[0160] Spacer **5240** of FIGS. **69-73** is similar to the embodiment of FIGS. **65-68**, with certain distinctions, described herein. Elements having a similar function, therefore, will be similarly numbered, and reference may be had to the foregoing discussion for a description of their structure and function.

[0161] Segments **5242**, **5244** form a sliding connection therebetween formed by interlocking projections **5246** and recesses **5248**. A variety of interlocking forms may be produced by interlocking projection **5246** and recess **5248**, including for example a mortise and tenon, dovetail, and half-dovetail type sliding joint. If a pair of interlocking projections **5246**, **5248** are provided, as illustrated, they may be configured to cooperate to form a secure sliding connection. For example, each projection may be provided with a single angular face, on relatively opposing sides of each of two projections **5246**.

[0162] As may be seen in FIGS. **72-73**, in spacer **5240**A, projection **5246** and recess **5248**, of segments **5242**A, **5244**A, are oriented to be slidable along an axis that is transverse with respect to an anterior-posterior axis of the body, when extensions **5212** are secured to an anterior face of vertebrae **300**, **302**. In this manner, alignment of the vertebrae may be carried out along the coronal plane, as opposed to along the sagittal plane, as illustrated for spacers **5200** and **5240**, when the respective spacer is implanted and affixed to the body. Projection **5246** and recess **5248** may thus be aligned along any axis of the body, to address a particular pathology.

[0163] Once a therapeutic alignment has been achieved, segments **5242**A, **5244**A are advantageously secured in relation to each other, to preserve an intended alignment of connected bones. As may be seen in FIG. **72**, an optional set screw **5250** may be used to secure segments **5242**, **5244**, or any segmented device herein, or alternatively, other fasteners of the disclosure may be used, such fasteners including adhesive. Further, ramped engagement surfaces **5218**, **5220**, shown in FIGS. **65-68**, may be provided. In addition, or in the alternative, other bone fixation devices may be used in combination with spacer **5240**, **5240**A, or any other device of the disclosure.

[0164] It should be understood that either segment **5242** or **5244** may be formed with a projection **5246** or recess **5248**, with the mating segment assuming the alternate profile.

[0165] Referring now to FIGS. **74-78**, a spacer **5260** includes a coupling formed by a groove **5262** into which a complementary projection **468** of plate **466** is inserted. Groove **5262** and projection **468** may be shaped in any of a variety of mating configurations, including, as examples, a mortise and tenon, or a dovetail. It should be noted that foregoing coupling may be provided for other assemblies of a spacer and plate, in this disclosure; likewise, spacer **5260** and plate **466** may be connected using alternate methods of the disclosure.

[0166] In an embodiment, spacer **5260** and plate **466** are inserted into a space between bones of a joint, for example in the intervertebral space, until a plate extension **470** contacts bone of the joint, for example bone of vertebra **302**. In the embodiment of FIG. **74**, plate extension **470** has a projecting shape **472** sized and dimensioned to contact a surface of the sacrum **310**, as shown in FIG. **79**, although projecting shape **472** may be adapted for contacting a different surface of the body, as shown for example in plate **466**A of FIG. **75**.

[0167] In FIGS. **76-77**, it may be seen that bone screws **200**, passed through plate apertures **474** of plate **466**, may form angles along their longitudinal dimension that mutually converge. This is advantageous, for example, to ensure that screws **200** obtain purchase in supportive tissue, that they do not extend vertebral body, and or that they do not interfere with body tissue that must be protected. In FIGS. **74-75** and **78**, it may be seen that screws **200** may also form divergent angles. Plate apertures **474** may be formed to guide screws **200** along intended angles, or may be polyaxial, whereby the medical practitioner may choose a therapeutically effective angular disposition of screw **200**, or other fastener.

[0168] FIGS. **74-79** further illustrate that implanted devices of the disclosure may be formed with chamfered edges **476**, to facilitate implantation, and to reduce interference with body tissue. Additionally, where a screw or other fastener passes through a device of the disclosure, it is advantageous to ensure that the fastener does not back out from engagement with the device or body tissue, which could lead to injury or failure of the device. FIG. **78** illustrates a locking screw or pin **442** operative to prevent substantially movement of screw **200**, or other elongated fastener, in a direction reverse to an installation direction, after implantation. A reduced profile portion **478** of pin **442** admits passage of screw **200**, when pin **442** is rotated so that reduced profile **478** faces aperture **474**, as shown in a lower left screw of FIG. **78**. After a head portion of screw **200** has passed reduced profile **478**, pin **442** may be rotated so that a non-reduced portion of pin **442** blocks a path of screw **200**, as shown for the lower right screw.

[0169] FIGS. **80-84** illustrate tool **6020**A, a variation of tool **6020** of FIGS. **16-18**, illustrated in use together with the embodiment of FIG. **74**. In FIG. **80**, it may be seen that tool **6020**A includes handles, advantageously including a biasing element **6102**, and a lock **6104** to maintain handles in a desired disposition. A pivot **6106** enables movement of handles to effect a corresponding movement of grippers **6108**, shaped and dimensioned to cooperate with tool engagement surfaces **5178**, which are advantageously formed upon spacer **5260** and or plate **466**, for this purpose.

[0170] The functions of engagement surface **6030**, push rod **6026**, and guide **6028** have been explained with respect to FIGS. **16-18**. It may further be seen that push rod **6026** is formed with a groove **6110** engageable with a projection of guide **6028**, operative to thereby prevent rotation of push rod **6026** and connected engagement surface **6030**. Additionally visible is an internally threaded collar **6112**, cooperative with a threaded external portion **6114** of push rod **6026**, threaded collar **6112** rotatably fixed within support **6116** to advance or withdraw push rod **6026** to effectuate movement of contacted body tissue. A proximal tool engaging end **6118** of threaded collar **6112** may be engaged and driven by a tool, for example a powered drill or t-handle, to rotate collar **6112**. As rotation and reduction take place along the same axis, control of the reduction is optimized.

[0171] In FIGS. **16-18**, a threaded connection **6024** is formed between tool **6020** and spacer **5030**. In the embodiment of FIGS. **80-84**, grippers **6108** engage an external surface of spacer **5260**. Alternatively, both grippers **6108** and threaded connection **6024** may be formed. FIG. **81** illustrates an assembly of spacer **5260** and plate **466** gripped by tool **6020**A. In FIG. **82**, tool **6020**A has been used to push the assembly into a disc space formed between vertebra **300** and sacrum **310**. Subsequently, screws **200** have been inserted into the caudal vertebra, or sacrum **310**. In FIG. **83**, engagement surface **6030** has been advanced, through rotation of threaded collar **6112**, to engage and urge vertebrae **300** posteriorly with respect to sacrum **310**, into therapeutic alignment. In FIG. **84**, one or more screws **200** are inserted into the cephalad bone, vertebra **300**, to thereby fix an alignment of vertebra **300** and sacrum **310**.

[0172] FIGS. **85**A-**85**C depict a method of inserting a spacer in accordance with the disclosure. In FIG. **85**A, adjacent vertebrae **300**, **302** are being distracted. In some embodiments, the vertebrae are overdistracted to provide space for a spacer, such as any of those shown in the above embodiments. In FIG. **85**B, after overdistracting the vertebrae, vertebra **302** can be aligned with vertebra **300**. Any of the devices, systems and methods discussed above can be used to align the upper vertebra **302** with the lower vertebra **300**. In FIG. **85**C, after alignment of the vertebrae, a spacer **5000** with surface protrusions can be inserted into the disc space between the overdistracted vertebrae. Once the spacer is inserted into the disc space, the vertebrae **300**, **302** can be compressed over the spacer, thereby completing the surgical procedure.

[0173] FIG. **86** depicts a side view of a combinable spacer **5040**, plate having an upper lip **5055** and fasteners **202** of the disclosure. Advantageously, the upper lip **5055** of the plate can contact the upper vertebra **300**, such that when the combined spacer and plate is forced anteriorly, the upper lip **5055** can help to push the upper vertebra **300** into alignment with the lower vertebra.

[0174] FIG. **87** depicts a side view of an alternative plate and spacer combination, similar to that shown in FIG. **34**. In this embodiment, one or more screws **200** can be provided through the plate **420**. The screws **200** can be aligned such that at least one enters a vertebral body, while at least one other enters into a disc **306**.

[0175] The devices and methods of the disclosure enable delivering and implant and reducing spondylolisthesis from an anterior approach, although other approaches are possible. More particularly, the disclosure provides for delivering an implant and reducing the spondylolisthesis using a single tool, thereby at least reducing separate approaches into the body, and reducing time required for the procedure, for the benefit of the patient and the medical practitioners, while reducing risks and costs. Additionally, by performing insertion and reduction using a single implant, and by using a single tool, accuracy is improved, and manipulation of body tissue is minimized.

[0176] Spacers, plates, and fasteners of the disclosure may be formed with biocompatible materials, of sufficient purity, including for example PEEK (polyether ether ketone), titanium, stainless steel, or a cobalt chromium alloy. Other polymers, metals, alloys, or composite materials may alternatively be used, as known in the art, or hereinafter developed. A radiopacifier may be added to devices of the disclosure to improve visibility under imaging. Devices of the invention may be formed by extrusion, milling, forging, casting, molding, or any other method advantageously used for the materials selected and the structure intended.

[0177] It should be understood that, in the various embodiments illustrated and described herein, an assembly may include any or all of a spacer, plate, tool, and or fasteners, and although all such elements may be shown in a particular illustration, for brevity and compactness, one or more such elements may be eliminated in a particular medical application, as would be understood by one skilled in the art. For example, an implant such as a spacer may not require separate fasteners. Alternatively, a disc **306** may be healthy, and not in need of replacement, or a spacer may be used without a plate.

[0178] All references cited herein are expressly incorporated by reference in their entirety. There are many different features to the present invention and it is contemplated that these features may be used together or separately. Unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. Thus, the invention should not be limited to any particular combination of features or to a particular application of the invention. Further, it should be understood that variations and modifications within the spirit and scope of the invention might occur to those skilled in the art to which the invention pertains. Accordingly, all expedient modifications readily attainable by one versed in the art from the disclosure set forth herein that are within the scope and spirit of the present invention are to be included as further embodiments of the present invention.

## Claims

**1**. A method for therapeutically aligning two adjacent bones on opposite sides of a joint space, comprising: distracting the two adjacent bones; providing a device including: a first segment having a mounting extension and an intervertebral spacer portion configured to be disposed in the joint space; a second segment configured to mate with the first segment, the second segment including an aperture; a first fastener configured to be received in the mounting extension; and a second fastener configured to be received in the aperture; mating the first segment with the second segment; inserting the intervertebral spacer portion into the joint space; securing a first fastener through the mounting extension and into one of the two adjacent bones; and securing a second fastener through the aperture and into one of the two adjacent bones.